MIL-B-48201(MU)
13 November 1972
SUPERSEDING
see 6.5

# MILITARY SPECIFICATION

# BREECH BLOCK AND PIN ASSEMBLY

## 1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of breech block assembly for M422 projectile.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-T-46035 --- Titanium Alloys, High Strength Wrought (For Critical Components)

FSC 1110



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STANDARDS

MILITARY

MIL-STD-105 --- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167 -- Ammunition Data Cards
MIL-STD-1168 -- Lot Numbering of Ammunition
MIL-STD-1169 -- Packaging, Packing and Marking for Shipments of Inert Ammunition Components

DRAWINGS

ORDNANCE CORPS

8787126 ------- Breech Block and Pin Assembly
8787127 ------- Block, Breech
ET-8787126 ---- Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The breech block and pin assembly shall comply with all requirements specified on the drawing (dwg) 8787126, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).



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3.3 Workmanship.

3.3.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.3.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.3.3 Burrs. Parts shall be free from burrs.

3.3.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.3.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted (see 6.4). When the contract requires first article inspection, the contractor shall submit one complete assembly (see dwg 8787126) and one set of parts for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests need not be conducted in order specified in Table I.



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Table I

First Article Inspection and Tests

| Inspection or Test | Dest | Non-Dest | Parts | Assy |
|---|---|---|---|---|
| 1. Classification of Characteristics (see 4.3.4.1) | | X | X | X |
| 2. Physical Properties | X | | X | X |

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with MIL-STD-1168. Changes in lot numbers shall be recorded on Ammunition Data Cards in accordance with MIL-STD-1167 and MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.



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| Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size |
|---|---|---|---|---|---|---|---|
| 1-72 | all | 140-141 | 124 | 256-259 | 177 | 541-549 | 230 |
| 73 | 72 | 142 | 125 | 260-262 | 178 | 550-559 | 231 |
| 74 | 73 | 143-144 | 126 | 263-265 | 179 | 560-569 | 232 |
| 75 | 74 | 145 | 127 | 266-268 | 180 | 570-579 | 233 |
| 76 | 75 | 146-147 | 128 | 269-271 | 181 | 580-590 | 234 |
| 77 | 76 | 148-149 | 129 | 272-275 | 182 | 591-601 | 235 |
| 78 | 77 | 150 | 130 | 276-278 | 183 | 602-613 | 236 |
| 79 | 78 | 151-152 | 131 | 279-282 | 184 | 614-625 | 237 |
| 80 | 79 | 153-154 | 132 | 283-285 | 185 | 626-637 | 238 |
| 81-82 | 80 | 155-156 | 133 | 286-289 | 186 | 638-649 | 239 |
| 83 | 81 | 157 | 134 | 290-293 | 187 | 650-662 | 240 |
| 84 | 82 | 158-159 | 135 | 294-296 | 188 | 663-676 | 241 |
| 85 | 83 | 160-161 | 136 | 297-300 | 189 | 677-690 | 242 |
| 86 | 84 | 162-163 | 137 | 301-304 | 190 | 691-704 | 243 |
| 87 | 85 | 164-165 | 138 | 305-308 | 191 | 705-719 | 244 |
| 88-89 | 86 | 166-167 | 139 | 309-312 | 192 | 720-735 | 245 |
| 90 | 87 | 168 | 140 | 313-316 | 193 | 736-751 | 246 |
| 91 | 88 | 169-170 | 141 | 317-320 | 194 | 752-768 | 247 |
| 92 | 89 | 171-172 | 142 | 321-324 | 195 | 769-785 | 248 |
| 93 | 90 | 173-174 | 143 | 325-329 | 196 | 786-804 | 249 |
| 94-95 | 91 | 175-176 | 144 | 330-333 | 197 | 805-823 | 250 |
| 96 | 92 | 177-178 | 145 | 334-338 | 198 | 824-842 | 251 |
| 97 | 93 | 179-180 | 146 | 339-342 | 199 | 843-863 | 252 |
| 98 | 94 | 181-182 | 147 | 343-347 | 200 | 864-884 | 253 |
| 99-100 | 95 | 183-184 | 148 | 348-352 | 201 | 885-907 | 254 |
| 101 | 96 | 185-187 | 149 | 353-356 | 202 | 908-930 | 255 |
| 102 | 97 | 188-189 | 150 | 357-361 | 203 | 931-955 | 256 |
| 103 | 98 | 190-191 | 151 | 362-366 | 204 | 956-981 | 257 |
| 104-105 | 99 | 192-193 | 152 | 367-372 | 205 | 982-1008 | 258 |
| 106 | 100 | 194-195 | 153 | 373-377 | 206 | 1009-1036 | 259 |
| 107 | 101 | 196-197 | 154 | 378-382 | 207 | 1037-1066 | 260 |
| 108-109 | 102 | 198-200 | 155 | 383-388 | 208 | 1067-1098 | 261 |
| 110 | 103 | 201-202 | 156 | 389-393 | 209 | 1099-1131 | 262 |
| 111 | 104 | 203-204 | 157 | 394-399 | 210 | 1132-1166 | 263 |
| 112-113 | 105 | 205-207 | 158 | 400-405 | 211 | 1167-1204 | 264 |
| 114 | 106 | 208-209 | 159 | 406-411 | 212 | 1205-1243 | 265 |
| 115 | 107 | 210-212 | 160 | 412-417 | 213 | 1244-1285 | 266 |
| 116-117 | 108 | 213-214 | 161 | 418-424 | 214 | 1286-1330 | 267 |
| 118 | 109 | 215-217 | 162 | 425-430 | 215 | 1331-1378 | 268 |
| 119 | 110 | 218-219 | 163 | 431-437 | 216 | 1379-1429 | 269 |
| 120-121 | 111 | 220-222 | 164 | 438-443 | 217 | 1430-1483 | 270 |
| 122 | 112 | 223-224 | 165 | 444-450 | 218 | 1484-1542 | 271 |
| 123-124 | 113 | 225-227 | 166 | 451-457 | 219 | 1543-1605 | 272 |
| 125 | 114 | 228-230 | 167 | 458-465 | 220 | 1606-1673 | 273 |
| 126-127 | 115 | 231-232 | 168 | 466-472 | 221 | 1674-1747 | 274 |
| 128 | 116 | 233-235 | 169 | 473-480 | 222 | 1748-1827 | 275 |
| 129-130 | 117 | 236-238 | 170 | 481-488 | 223 | 1828-1914 | 276 |
| 131 | 118 | 239-241 | 171 | 489-496 | 224 | 1915-2010 | 277 |
| 132-133 | 119 | 242-243 | 172 | 497-504 | 225 | 2011-2115 | 278 |
| 134 | 120 | 244-246 | 173 | 505-513 | 226 | 2116-2232 | 279 |
| 135-136 | 121 | 247-249 | 174 | 514-521 | 227 | 2233-2361 | 280 |
| 137 | 122 | 250-252 | 175 | 522-530 | 228 | over 2361 | 298 |
| 138-139 | 123 | 253-255 | 176 | 531-540 | 229 | | |

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Table III

Minor Characteristics

| Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size |
|---|---|---|---|---|---|---|---|
| 1-16 | all | 28-29 | 23 | 52-56 | 31 | 132-156 | 39 |
| 17 | 16 | 30-31 | 24 | 57-62 | 32 | 157-191 | 40 |
| 18 | 17 | 32-34 | 25 | 63-69 | 33 | 192-244 | 41 |
| 19-20 | 18 | 35-36 | 26 | 70-76 | 34 | 245-334 | 42 |
| 21 | 19 | 37-40 | 27 | 77-86 | 35 | 335-519 | 43 |
| 22-23 | 20 | 41-43 | 28 | 87-97 | 36 | 520-1120 | 44 |
| 24-25 | 21 | 44-47 | 29 | 98-112 | 37 | over 1120 | 45 |
| 26-27 | 22 | 48-51 | 30 | 113-131 | 38 | | |

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Block, breech (see dwg 8787127) covering a detail of dwg 8787126.

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | | |
| | 1. Flash hole present and not obstructed ..... | Visual |
| Major: | | |
| 101. | Concentricity of pitch diameter of internal thread and perpendicularity with internal thread cavity bottom surface ..... | Gage |
| 102. | Concentricity of outside diameter at right end ..... | Gage |
| 103. | Perpendicularity of front surface ..... | Gage |
| 104. | Location of flash hole large counterbore ..... | Gage |
| 105. | Location of cone cavity ..... | Gage |
| 106. | Location of large pin cavity ..... | Gage |



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| | | |
|---|---|---|
| 107. | Location of threaded cavities in right face | Gage |
| 108. | Location of small pin cavities in bottom of internal thread cavity | Gage |
| 109. | Pitch diameter of internal thread | Gage |
| 110. | Minor diameter of internal thread | Gage |
| 111. | Pitch diameter of external thread | Gage |
| 112. | Major diameter of external thread | Gage |
| 113. | Diameter of flash holes | Gage |
| 114. | Depth of internal threads, min. | Gage |
| 115. | Diameter at mouths of cone cavities | Gage |
| 116. | Diameter of flash hole counterbore | Gage |
| 117. | Diameter of angular flash hole counterbore | Gage |
| 118. | Threads not damaged | Gage |

Minor:

| | | |
|---|---|---|
| 201. | Concentricity of flash hole diameter | Gage |
| 202. | Diameter of small pin cavity | Gage |
| 203. | Outside diameter at left end | Gage |
| 204. | Length from right face to external thread | Gage |
| 205. | Depth of left cavity | Gage |
| 206. | Depth of angular flash hole counterbore | Gage |
| 207. | Depth of flash hole counterbore | Gage |
| 208. | Depth of large pin cavity | Gage |
| 209. | Depth of small pin cavity | Gage |
| 210. | Total length, min. | Gage |
| 211. | Depth of wrench slots | Gage |
| 212. | Width of wrench slots | Gage |
| 213. | Angle of cone cavity proper | Gage |
| 214. | Length from right surface to external thread | Gage |
| 215. | Length from left thread to external thread | Gage |
| 216. | Finish proper | Gage |
| 217. | Workmanship (see 3.4) | Visual |
| 218. | Feather edge removed from external thread | Visual |



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4.3.2.1.2 Breech block and pin assembly (see dwg 8787126).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: None defined | | |
| Major: | | |
| 101. | Extension of large pin not excessive | Gage |
| 102. | Extension of small pin not excessive | Gage |
| 103. | Assembly secure | Visual/Manual |
| Minor: | | |
| 201. | Stamping present, correct, and identifiable | Visual |
| 202. | Workmanship (see 3.4) | Visual/Manual |

4.3.3 Testing. This test shall be performed.

| Test | Test Classification |
|---|---|
| a. Physical properties | Major |

This test is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Physical properties (see dwg 8787127). Each breech block shall be tested in accordance with 4.4.1. If any sample fails to comply with MIL-T-46035, the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions and calibration instructions specified on the Equipment Lists (EL's). EL's for assemblies are listed on the Equipment Tabulation, ET-8787126 When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.



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a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Physical properties (see 4.3.3.1). Each breech block shall be tested in accordance with MIL-T-46035.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.4 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.



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## 6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1 and 5.4). Data specified in 4.3.4.1 and 5.4 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-N-I.

6.4 Instructions for first article inspection (see 4.2.1). Parts may be used in subassemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government. Subassemblies may be used in final assemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1612 (Rev 1), dated 15 February 1962.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1110-A033



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-B-48201(MU) Breech Block and Pin Assembly

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.



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